

1 Lecture 16: April 23, 2026

Lecture Overview:

1.1 The Gamma Function

Section Overview:

Recall the **Gamma function** $\Gamma(x)$. It is characterized by the following three properties:

- (i) $\Gamma(x+1) = x\Gamma(x)$,
- (ii) $\Gamma(1) = 1$,
- (iii) $\log \Gamma$ is convex.

Theorem 1.1. *These three properties uniquely define Γ .*

Proof. It suffices to show the result for all $x \in (0, 1)$, since property (i) then determines Γ on all of $(0, \infty)$.

If f is convex and $x_1 < x_2 < x_3$, then

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} \leq \frac{f(x_3) - f(x_1)}{x_3 - x_1} \leq \frac{f(x_3) - f(x_2)}{x_3 - x_2}.$$

Assume f is some function satisfying all three properties. Let $\varphi = \log f$. In particular, $f(n) = (n-1)!$ for all $n \geq 1$. Then

$$\varphi(n+1) - \varphi(n) = \log n! - \log(n-1)! = \log \frac{n!}{(n-1)!} = \log n,$$

and similarly,

$$\varphi(n+2) - \varphi(n+1) = \log(n+1).$$

For $x \in (0, 1)$, apply the convexity inequality with $x_1 = n$, $x_2 = n+x$, $x_3 = n+1$:

$$\frac{\varphi(n+x) - \varphi(n)}{x} \leq \frac{\varphi(n+1) - \varphi(n)}{1} = \log n.$$

Similarly, with $x_1 = n-1$, $x_2 = n$, $x_3 = n+x$:

$$\log(n-1) = \frac{\varphi(n) - \varphi(n-1)}{1} \leq \frac{\varphi(n+x) - \varphi(n)}{x}.$$

Combining:

$$x \log(n-1) \leq \varphi(n+x) - \varphi(n) \leq x \log n.$$

By property (i), $f(x+k+1) = (x+k)f(x+k)$, so applying this repeatedly:

$$\varphi(n+x) = \varphi(x) + \sum_{k=0}^{n-1} \log(x+k).$$

Since $\varphi(n) = \log(n-1)!$, we get

$$\log(n-1)! + x \log(n-1) - \sum_{k=0}^{n-1} \log(x+k) \leq \varphi(x) \leq \log(n-1)! + x \log n - \sum_{k=0}^{n-1} \log(x+k).$$

The difference between the upper and lower bounds is $x \log \frac{n}{n-1} \rightarrow 0$ as $n \rightarrow \infty$. By the squeeze theorem, $\varphi(x)$ is uniquely determined:

$$\varphi(x) = \lim_{n \rightarrow \infty} \left[\log(n-1)! + x \log n - \sum_{k=0}^{n-1} \log(x+k) \right],$$

so $f = e^\varphi$ is uniquely determined on $(0, 1)$, and hence on all of $(0, \infty)$ by property (i). $\square$

1.2 The Beta Function

Section Overview:

Definition 1.2. For $x > 0, y > 0$, we define the **Beta function** by

$$B(x, y) = \int_0^1 t^{x-1} (1-t)^{y-1} dt = \frac{\Gamma(x) \Gamma(y)}{\Gamma(x+y)}.$$

Proof. We will show that $f(x) = \frac{B(x, y) \Gamma(x+y)}{\Gamma(y)}$ satisfies the three properties, which by uniqueness proves equality.

Property (ii): $B(1, y) = \int_0^1 (1-t)^{y-1} dt = \frac{1}{y}$, and $\Gamma(1+y) = y \Gamma(y)$, so

$$f(1) = \frac{\frac{1}{y} \cdot y \Gamma(y)}{\Gamma(y)} = 1.$$

$\square$

Remark 1.3 (Probabilistic interpretation of Γ). 1. $\Gamma(k+1) = \int_0^\infty x^k e^{-x} dx = E[X^k]$ is the k -th moment of an exponential random variable.

2. The Gamma distribution models the waiting time until the k -th event in a Poisson process.

1.3 Stirling's Formula

Section Overview:

Theorem 1.4 (Stirling's Approximation).

$$n! \sim \sqrt{2\pi n} \left(\frac{n}{e}\right)^n \quad \text{as } n \rightarrow \infty.$$

Theorem 1.5 (Stirling's Limit).

$$\lim_{n \rightarrow \infty} \frac{n!}{\sqrt{2\pi n} \left(\frac{n}{e}\right)^n} = 1.$$

Proof. Starting from $\Gamma(n+1) = n! = \int_0^\infty t^n e^{-t} dt$, substitute $t = n(1+u)$:

$$n! = n^{n+1} e^{-n} \int_{-1}^\infty (1+u)^n e^{-nu} du = n^{n+1} e^{-n} \int_{-1}^\infty e^{n\psi(u)} du,$$

where $\psi(u) = \log(1+u) - u$. Note that $\psi(0) = 0$, $\psi'(0) = 0$, $\psi''(0) = -1$, and ψ has a unique global maximum at $u = 0$.

Substitute $u = s/\sqrt{n}$:

$$n! = \frac{n^{n+1/2}}{e^n} \int_{-\sqrt{n}}^\infty e^{n\psi(s/\sqrt{n})} ds.$$

Since $\psi(u) = -u^2/2 + O(u^3)$ near $u = 0$, for fixed s :

$$n\psi\left(\frac{s}{\sqrt{n}}\right) = n\left(-\frac{s^2}{2n} + O\left(\frac{s^3}{n^{3/2}}\right)\right) = -\frac{s^2}{2} + O\left(\frac{s^3}{\sqrt{n}}\right) \rightarrow -\frac{s^2}{2}.$$

The integrand $e^{n\psi(s/\sqrt{n})}$ converges pointwise to $e^{-s^2/2}$ and is dominated by an integrable function (since $\psi(u) < 0$ for all $u \neq 0$ and $\psi(u) \sim -u^2/2$). By dominated convergence,

$$\int_{-\sqrt{n}}^\infty e^{n\psi(s/\sqrt{n})} ds \rightarrow \int_{-\infty}^\infty e^{-s^2/2} ds = \sqrt{2\pi}.$$

Therefore

$$n! \sim \frac{n^{n+1/2}}{e^n} \cdot \sqrt{2\pi} = \sqrt{2\pi n} \left(\frac{n}{e}\right)^n. \quad \square$$

1.4 Introduction to Multivariable Differentiation

Section Overview:

In the book, we write $F' = f$, $G' = g$. Then

$$\int fG + \int gF = FG.$$

Let V be a vector space over $\mathbb{R}$.

Definition 1.6. A function $f : V^k \rightarrow \mathbb{R}$ is **k -linear** (or **multilinear**) if it is linear in each coordinate:

$$f(\dots, au + bv, \dots) = a f(\dots, u, \dots) + b f(\dots, v, \dots).$$

Such a function is also called a **k -tensor**, where k is the **degree**.

There is a high-dimensional tangent plane to every surface. Given a “nice shape” (manifold), every point p has a **tangent vector space** T_p . In particular,

$$T_p(\mathbb{R}^n) \cong \mathbb{R}^n.$$

“Nice” means all tangent spaces T_p have the same dimension.

Definition 1.7. A **1-form** on X is a map that assigns to each $p \in X$ a linear function on $T_p(X)$.

Example. On $\mathbb{R}^2$, dx and dy are 1-forms: dx picks out the first coordinate of a tangent vector, and dy picks out the second.

Properties of Tensor Products